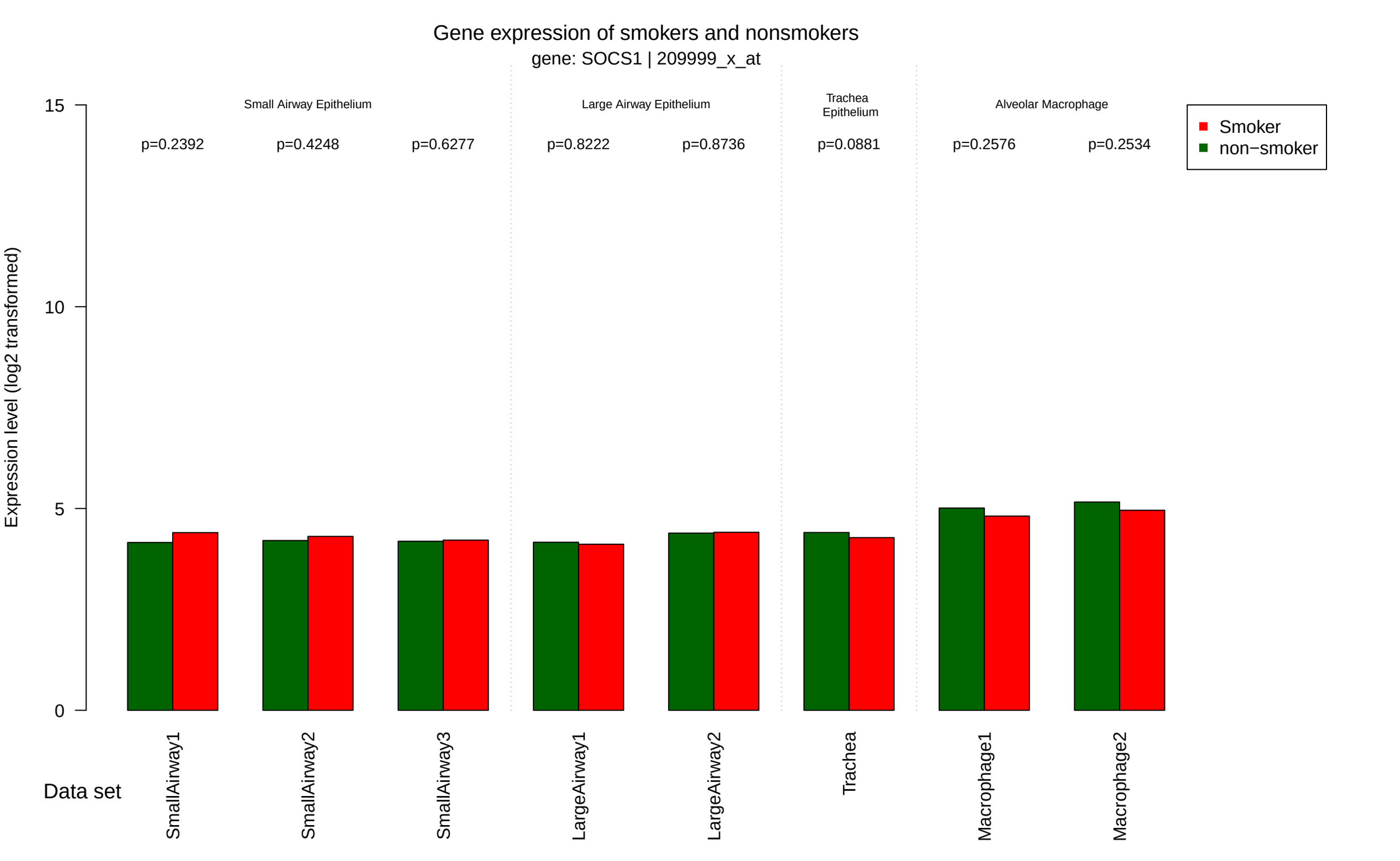
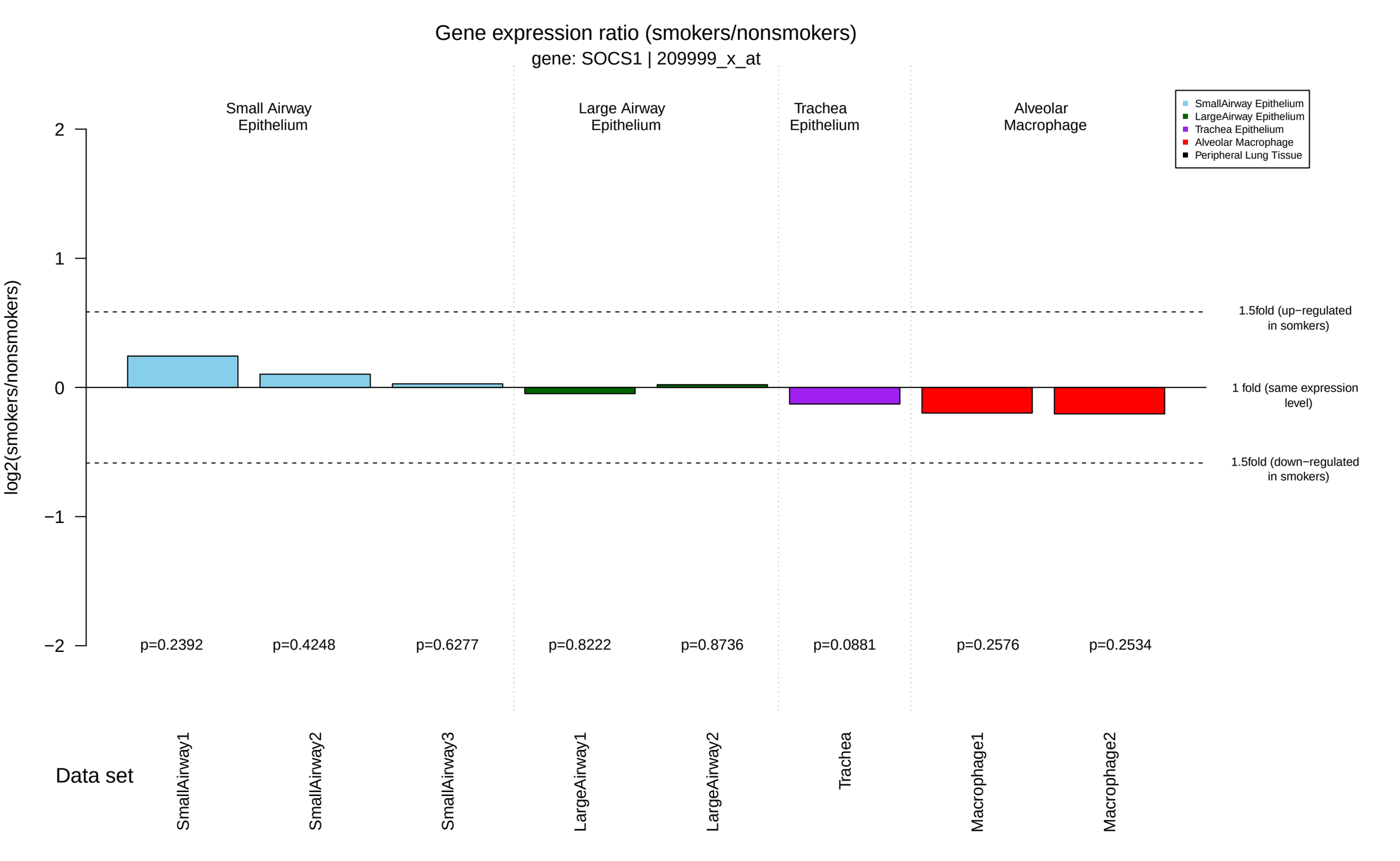
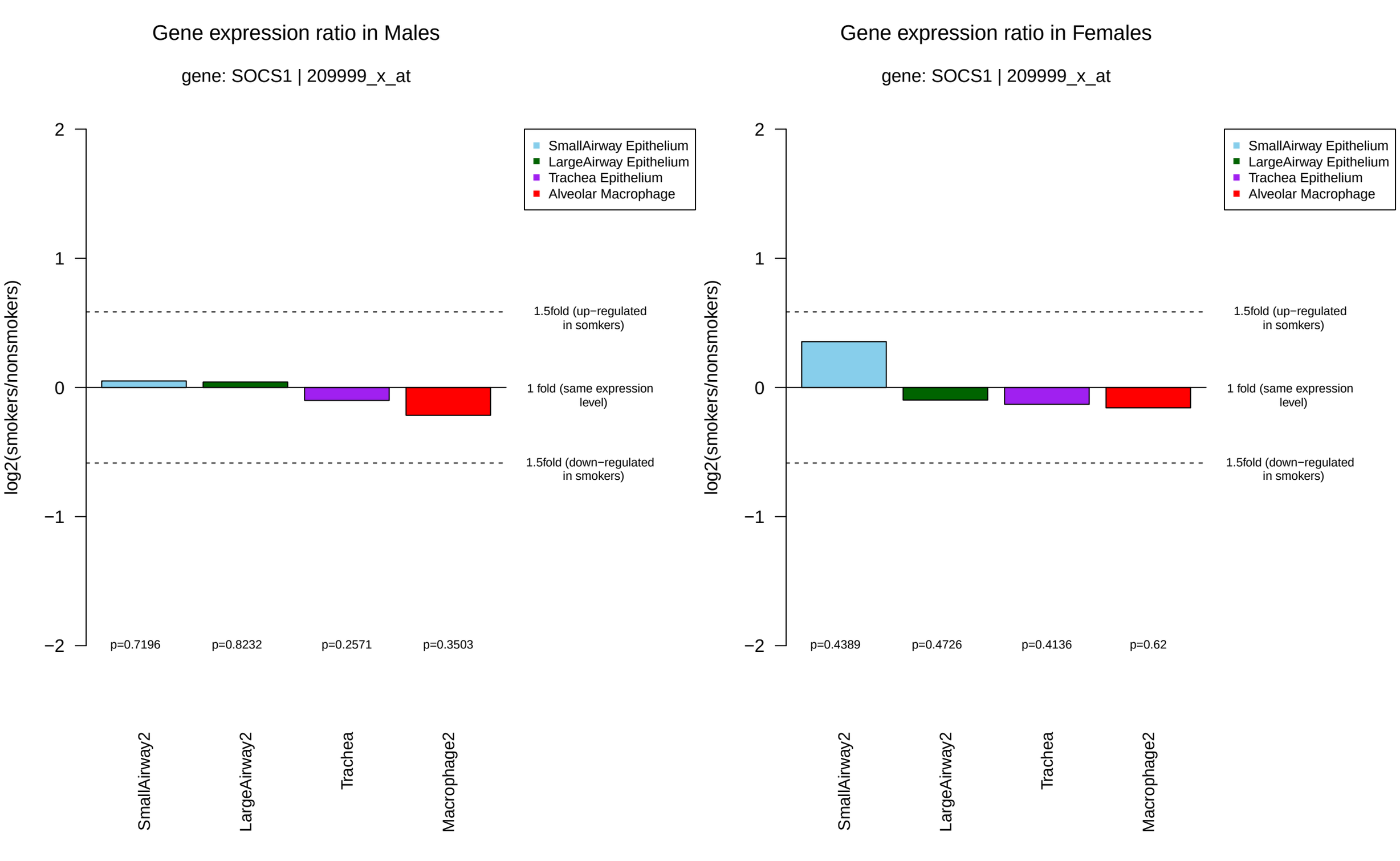
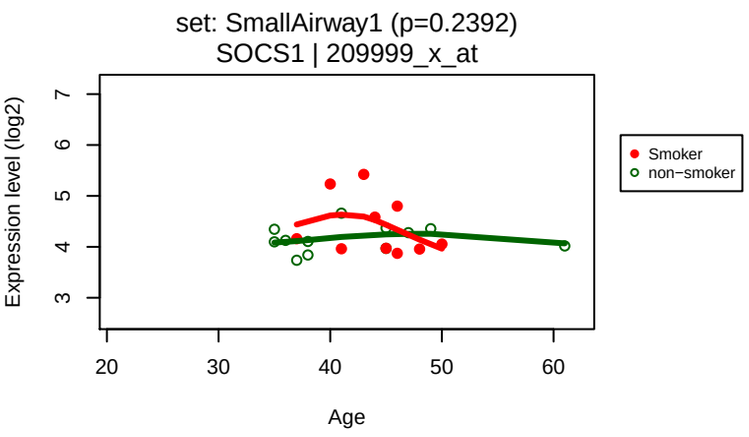
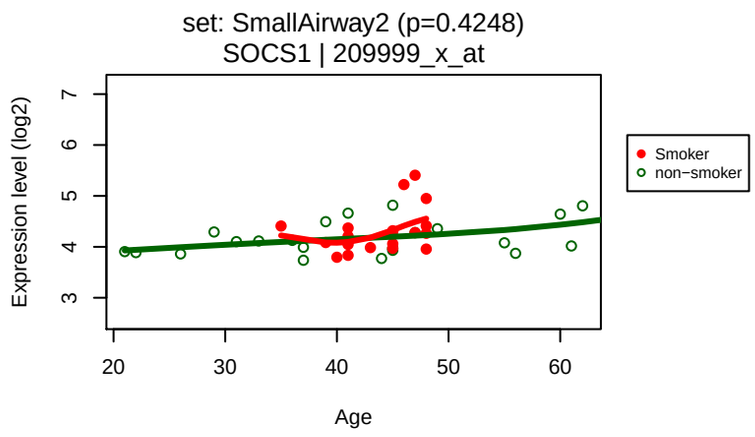
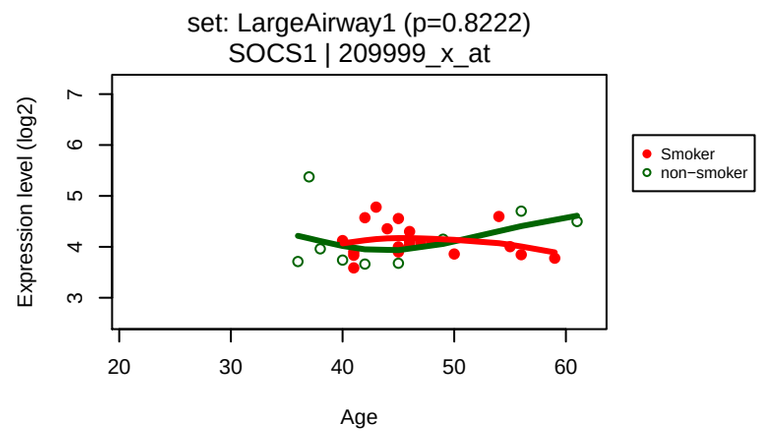

Gene expression of smokers and nonsmokers
gene: SOCS1 | 209999_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2392
p=0.4248
p=0.6277
p=0.8222
p=0.8736
p=0.0881
p=0.2576
p=0.2534
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

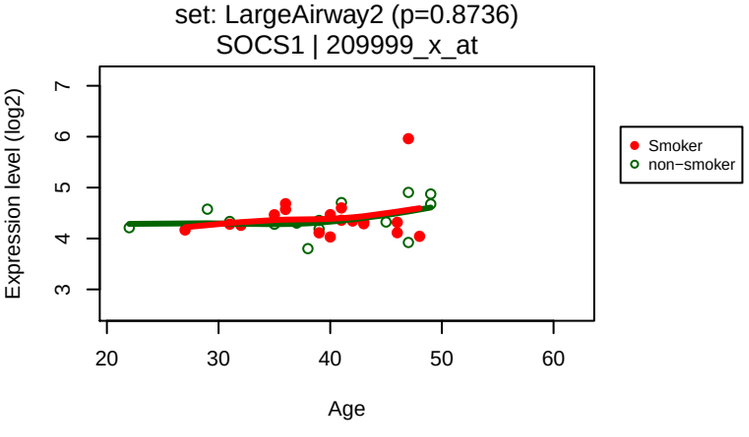

Gene expression ratio (smokers/nonsmokers)
gene: SOCS1 | 209999_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2392
p=0.4248
p=0.6277
p=0.8222
p=0.8736
p=0.0881
p=0.2576
p=0.2534
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

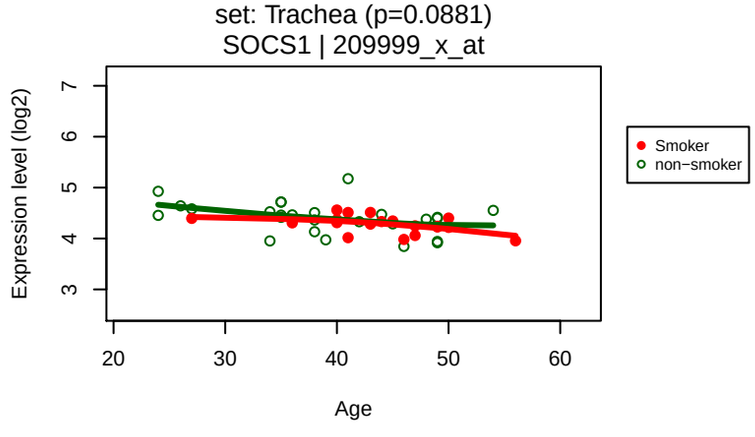

Gene expression ratio in Males
gene: SOCS1 | 209999_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7196
p=0.8232
p=0.2571
p=0.3503
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SOCS1 | 209999_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4389
p=0.4726
p=0.4136
p=0.62
SmallAirway2
LargeAirway2
Trachea
Macrophage2

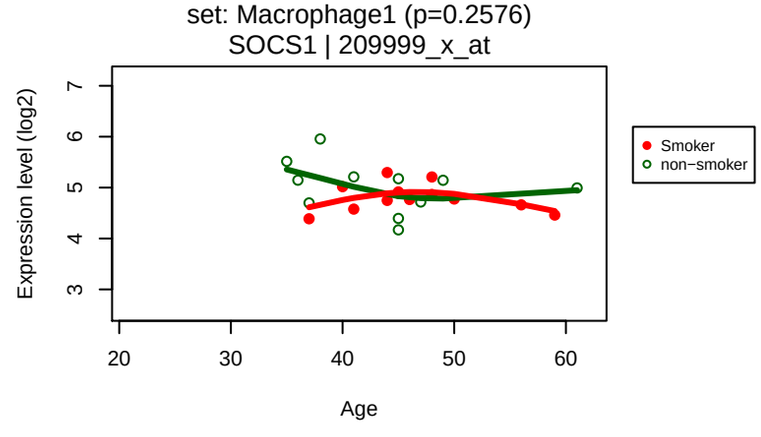

set: SmallAirway1 (p=0.2392)
SOCS1 | 209999_x_at
Expression level (log2)
Age
Smoker
non−smoker

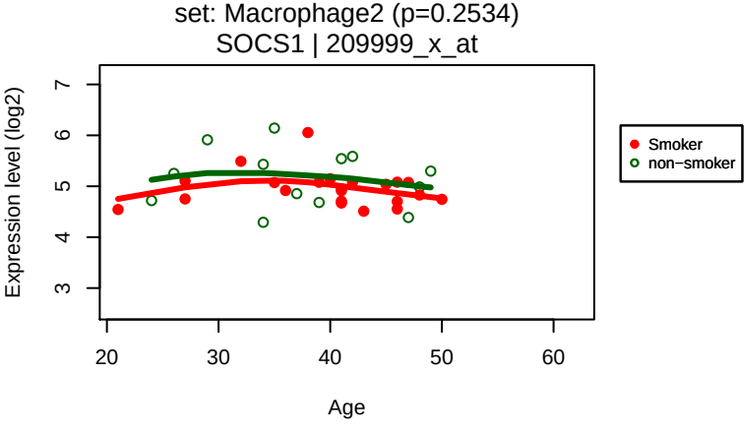

set: SmallAirway2 (p=0.4248)
SOCS1 | 209999_x_at
Expression level (log2)
Age
Smoker
non−smoker

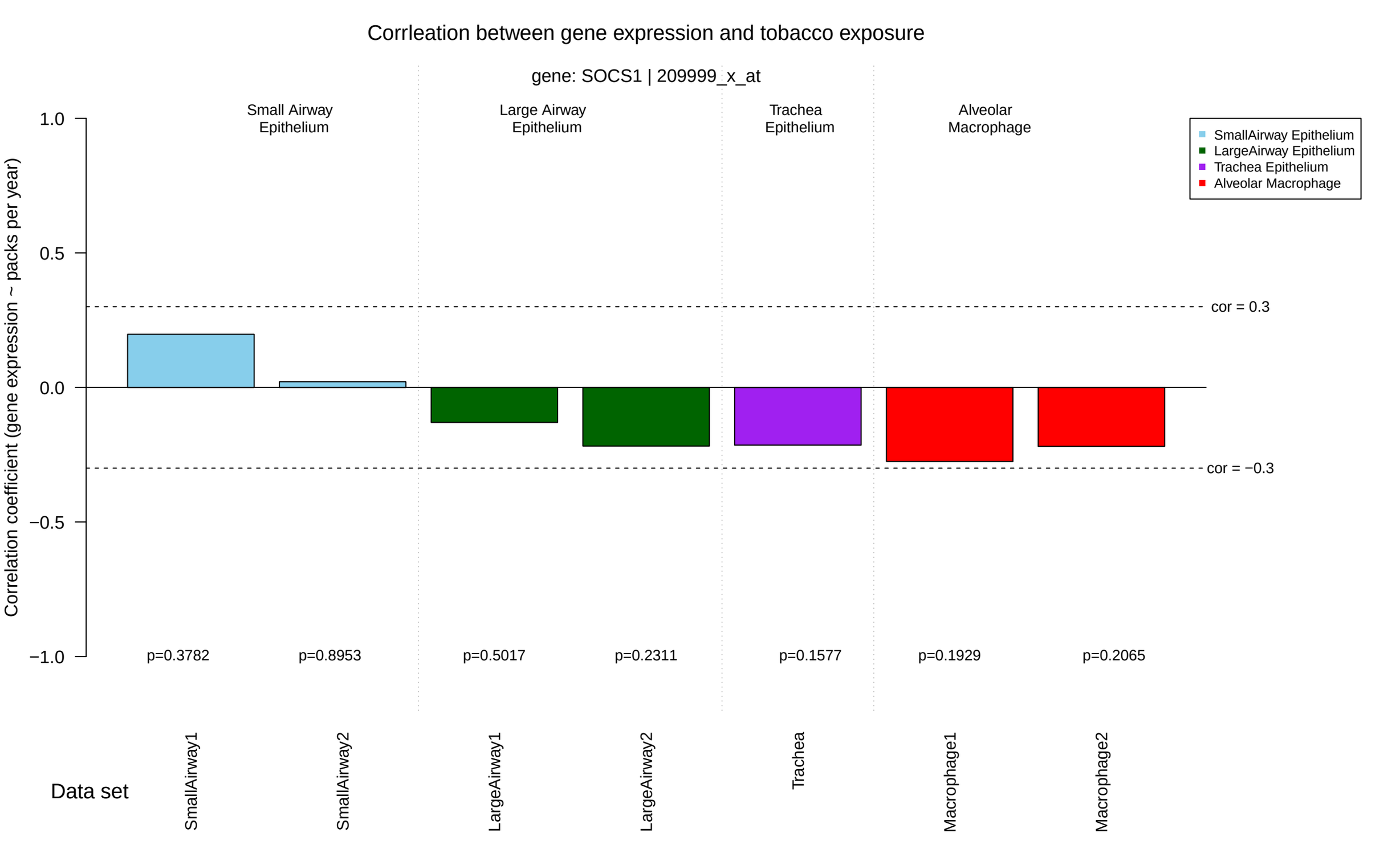

set: LargeAirway1 (p=0.8222)
SOCS1 | 209999_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8736)
SOCS1 | 209999_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0881)
SOCS1 | 209999_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2576)
SOCS1 | 209999_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2534)
SOCS1 | 209999_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SOCS1 | 209999_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3782
p=0.8953
p=0.5017
p=0.2311
p=0.1577
p=0.1929
p=0.2065
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2